ASSIGNMENT REPORT

INSTALLATION AND CONFIGURATION OF A DEBIAN VIRTUAL MACHINE

V Semester | CSS First Assignment

OPERATING SYSTEM

Debian GNU/Linux 10.11.0 (64-bit)

SUBMITTED BY

Vikram Aditya Kumar Suman

Roll Number: 24BECCS60

Virtualization Platform: Oracle VirtualBox

Submission date: 15 October 2026

1. Objective

To use Oracle VirtualBox to load and run a pre-configured virtual machine image, verify that the assigned operating system starts successfully, and document the virtual machine configuration and setup process.

2. Assigned Operating System and Software

Student name	Vikram Aditya Kumar Suman
University roll number	24BECCS60
Assigned operating system	Debian 10 (version 10.11.0), 64-bit
Virtualization software	Oracle VirtualBox
Virtual machine image	Pre-configured VirtualBox image from Linux VM Images

3. Virtual Machine Configuration

Setting	Configured value
Guest operating system	Debian (64-bit)
Allocated base memory (RAM)	2048 MB (2 GB)
Virtual processors	2
Virtual disk	512.00 GB virtual capacity; VDI, dynamically allocated
Network adapter	Intel PRO/1000 MT Desktop - NAT
Display setting used for troubleshooting	128 MB video memory; 3D acceleration disabled; VMSVGA selected if available

4. Setup Procedure

1. Installed and opened Oracle VirtualBox on the Windows host computer.
2. Downloaded and extracted the pre-configured Debian 10.11.0 VirtualBox image assigned to table entry 60 for university roll number 24BECCS60.
3. Opened the VirtualBox machine-definition (.vbox) file and ensured the virtual disk (.vdi) was attached from the extracted folder.
4. Reviewed the virtual machine configuration: Debian 64-bit, 2 GB RAM, 2 CPU processors, and a dynamically allocated 512 GB virtual disk.
5. Adjusted the display settings to troubleshoot an initial black screen, including increasing video memory and disabling 3D acceleration.
6. Started the virtual machine, accessed the Debian text terminal, and launched the graphical GNOME desktop. The desktop became responsive to mouse input.

5. VirtualBox Configuration Evidence

The screenshot below records the VM selected in Oracle VirtualBox Manager. It shows the guest OS type, RAM allocation, processor count, and virtual disk capacity.

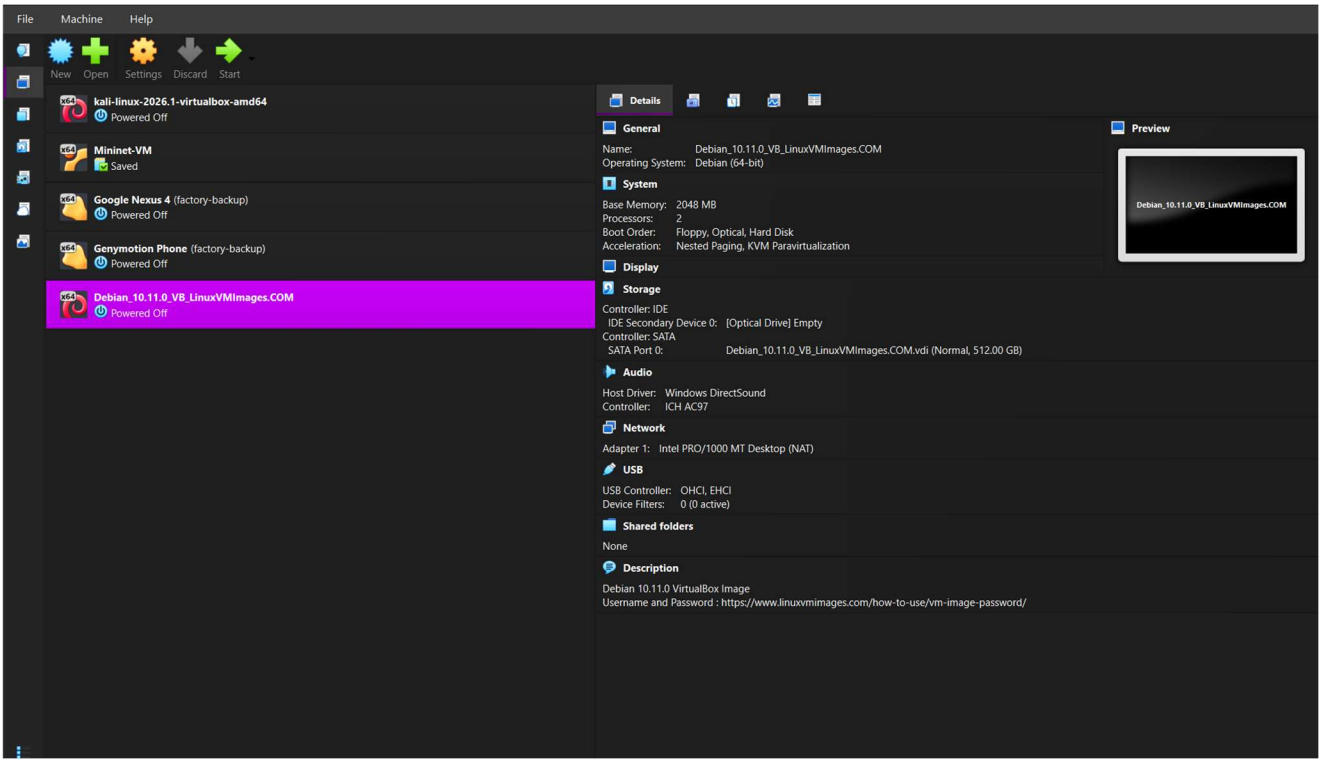


Figure 1. Oracle VirtualBox Manager - Debian 10.11.0 configuration.

Configuration values visible in Figure 1

Operating system: Debian (64-bit) | Base memory: 2048 MB | Processors: 2 | Virtual disk: 512.00 GB | Network: NAT

The 512 GB figure is the virtual capacity presented to the guest. The disk is dynamically allocated, so the host does not reserve the full virtual capacity immediately.

6. Performance and Troubleshooting Observations

At first, the VM displayed a black screen after boot. The text console was accessible, and the graphical desktop appeared after display troubleshooting and launching the graphical session. The GNOME desktop was responsive to mouse interaction. Performance was assessed only for basic startup and desktop use; demanding workloads were not tested.

7. Operating System Running Evidence

This screenshot shows the Debian GNOME graphical desktop after the virtual machine booted successfully.



Figure 2. Debian 10.11.0 GNOME desktop running in VirtualBox.

8. Network and Additional Software

Networking is configured through VirtualBox NAT. Internet access was not independently verified during this exercise. No additional software packages were reported as installed after launching the VM.

9. Screen Recording Link

Recording link: https://drive.google.com/file/d/1lsGQYaDARxMcvlCUHV7CmFFUVhmmWYqm/view?usp=drive_link

10. Conclusion

This exercise provided practical experience with loading a pre-configured Linux virtual machine in Oracle VirtualBox, checking its RAM, processor, storage, and network settings, and starting the Debian GNOME desktop. Troubleshooting the initial display issue also demonstrated the importance of VM graphics settings and keeping the virtual machine definition and disk files available at the expected paths.